Component	Pin Function	ESP32 GPIO	Pin Type	Hardware Action / Notes
All ToF Sensors	I2C SDA	GPIO 21	Safe	Connect all 3 sensor SDA pins here.
All ToF Sensors	I2C SCL	GPIO 22	Safe	Connect all 3 sensor SCL pins here.
Left ToF	XSHUT	GPIO 13	Safe	Pull HIGH in code to activate.
Front ToF	XSHUT	GPIO 14	Safe	Pull HIGH in code to activate.
Right ToF	XSHUT	GPIO 27	Safe	Pull HIGH in code to activate.
Servo 1	PWM Signal	GPIO 25	Safe	Assign to ESP32 PWM Channel in code.
Servo 2	PWM Signal	GPIO 26	Safe	Assign to ESP32 PWM Channel in code.
Servo 3	PWM Signal	GPIO 32	Safe	Assign to ESP32 PWM Channel in code.
DC Motor 1	Speed (PWM)	GPIO 33	Safe	Connect to Motor Driver Enable pin.
DC Motor 1	Direction (DIR)	GPIO 19	Safe	Connect to IN1 (and NOT gate to IN2).
DC Motor 2	Speed (PWM)	GPIO 4	Safe	Connect to Motor Driver Enable pin.
DC Motor 2	Direction (DIR)	GPIO 23	Safe	Connect to IN1 (and NOT gate to IN2).
DC Motor 3	Speed (PWM)	GPIO 16	Safe	Connect to Motor Driver Enable pin.
DC Motor 3	Direction (DIR)	**GPIO 5**	**Strapping**	⚠️ **Must add a 10kΩ Pull-Up resistor to 3.3V.**
DC Motor 4	Speed (PWM)	GPIO 17	Safe	Connect to Motor Driver Enable pin.
DC Motor 4	Direction (DIR)	**GPIO 15**	**Strapping**	⚠️ **Must add a 10kΩ Pull-Up resistor to 3.3V.**
DC Motor 5	Speed (PWM)	GPIO 18	Safe	Connect to Motor Driver Enable pin.
DC Motor 5	Direction (DIR)	**GPIO 2**	**Strapping**	⚠️ **Must add a 10kΩ Pull-Down resistor to GND.**